



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECV2223-03 (Web Programming)

Semester 02, 2025/2026

Assignment 2: Personal Portfolio Website with HTML, CSS, and JavaScript

Lecturer:

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Students:

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Reflection on Design Process and Web Optimization

Objective & Design Strategy

The goal of this assignment was to develop a responsive personal portfolio using only HTML, CSS, and vanilla JavaScript. As a Computer Networking and Security student, I prioritized a clean, accessible architecture over heavy frameworks. Visually, I adopted a dark, minimalist aesthetic to ensure optimal text contrast and project a professional technical persona.

Technical Implementation

To build a solid foundation, I utilized semantic HTML5 tags (<header>, <main>, <footer>). For the layout, I combined CSS Flexbox for the navigation bar to ensure symmetrical alignment, and CSS Grid (grid-template-columns: repeat(auto-fit...)) in the Portfolio section. This allowed project cards (e.g., MP4 Downloader, Video Compressor) to dynamically resize across devices. For interactivity, I used vanilla JavaScript to implement a responsive mobile menu, a dynamic time-based greeting, and a typewriter effect using setInterval(). Furthermore, I secured the global configuration object to prevent Prototype Pollution vulnerabilities.

Lighthouse Audit & Improvements

Following the initial build, a Google Lighthouse audit guided several strategic optimizations, resulting in exceptional scores (Performance: 99, Accessibility: 95, Best Practices: 100, SEO: 100):

- **Performance (99):** To prevent Cumulative Layout Shift (CLS) during rendering, I explicitly defined width and height attributes for the profile image. I also applied loading="lazy" to all project images to optimize network resource consumption.
- **Accessibility (95):** I enhanced screen-reader compatibility by adding aria-label attributes to the mobile menu and social links, and utilized aria-current="page" to indicate active navigation states.
- **Best Practices & SEO (100):** I fortified the site's security by appending rel="noopener noreferrer" to all external anchor tags, mitigating reverse tabnabbing risks. Comprehensive meta description tags were also added to guarantee full SEO compliance.

Conclusion

This project reinforced the power of core web technologies. By strictly prioritizing semantic structure, modern CSS layouts, and rigorous Lighthouse auditing, I successfully delivered a highly optimized and secure web portfolio.